

1.INTRODUCTION

1.1 Basics of Project

In recent years, Artificial Intelligence (AI) has emerged as a transformative technology with applications spanning various domains, including waste management. One innovative application of AI is in creating smart recycling systems that can automatically sort recyclable materials from non-recyclables. In this essay, we will delve into the basics of developing a project centered around an AI-powered video classifying recycle bin.

Understanding the Project:

The fundamental concept of the project revolves around creating a system that can analyze video footage of waste items being disposed of and classify them into recyclable and non-recyclable categories. The core components of this project include:

1. **Video Input:** The project requires a video input source, such as a camera, to capture footage of the waste items being discarded into the recycle bin. This footage serves as the input for the AI model.
2. **AI Model:** The heart of the project lies in the AI model, which is trained to recognize and classify different types of waste items. Convolutional Neural Networks (CNNs) are commonly used for image and video classification tasks in AI projects. The AI model must be trained on a diverse dataset containing images or videos of recyclable and non-recyclable items.
3. **Classification Algorithm:** The AI model employs a classification algorithm to categorize each frame of the video footage into recyclable or non-recyclable classes. This algorithm utilizes the learned features from the training data to make predictions on unseen waste items.
4. **User Interface:** To provide a user-friendly experience, a graphical user interface (GUI) can be developed to display the real-time video feed from the camera along with the classification results. This interface allows users to monitor the recycling process and ensures transparency.

Development Process:

Developing a project on AI video classifying recycle bin involves several key steps:

1. **Data Collection:** The first step is to gather a diverse dataset of waste items. This dataset should include images or videos of various recyclable and non-recyclable items commonly found in waste streams. Proper labeling of the data is essential for training the AI model.
2. **Preprocessing:** The collected data undergoes preprocessing steps such as resizing, normalization, and augmentation to enhance the quality and diversity of the dataset. Preprocessing helps in improving the generalization ability of the AI model.
3. **Model Training:** Using a deep learning framework such as TensorFlow or PyTorch, the AI model is trained on the preprocessed dataset. During training, the model learns to extract relevant features from the input video frames and make accurate predictions.
4. **Evaluation and Testing:** Once the model is trained, it is evaluated using a separate test dataset to assess its performance metrics such as accuracy, precision, and recall. Testing involves feeding unseen video footage to the model and analyzing its classification results.
5. **Integration:** The trained AI model is integrated into the project alongside the video input source and user interface components. Integration ensures that the system functions seamlessly, processing real-time video streams and providing accurate classification results.
6. **Deployment:** The final step involves deploying the project in a real-world setting, such as a recycling facility or public space. Continuous monitoring and maintenance are necessary to ensure the proper functioning of the AI-powered recycle bin.

In conclusion, developing a project on AI video classifying recycle bin involves combining elements of computer vision, machine learning, and user interface design to create an intelligent recycling system. By leveraging AI technology, we can streamline the waste management process, promote recycling practices, and contribute to environmental sustainability. As advancements in AI continue, the potential for innovative applications in waste management and beyond will only grow, paving the way for a smarter and more efficient future.

1.2 Scope of the Project

1. System Design and Architecture: The project scope encompasses designing the overall system architecture, including the hardware components (such as cameras and processing units), software modules (including AI model, classification algorithms, and user interface), and the integration between these components.

2. Data Collection and Annotation: Gathering a diverse dataset of waste items is crucial for training the AI model. The scope involves collecting, annotating, and preprocessing the dataset to ensure its quality and relevance to the task of waste classification.

3. AI Model Development: Developing the AI model involves selecting appropriate deep learning architectures (such as CNNs), training algorithms, and optimization techniques. The scope includes experimenting with different model architectures, hyperparameters, and training strategies to achieve optimal classification performance.

4. Training and Evaluation: Training the AI model on the collected dataset and evaluating its performance are integral parts of the project scope. This includes partitioning the dataset into training, validation, and test sets, monitoring the training process, and evaluating the model's accuracy, precision, recall, and other performance metrics.

5. Real-time Video Processing: Implementing real-time video processing capabilities is essential for the project's success. The scope involves developing algorithms and techniques to efficiently process video streams from the input source (e.g., camera feed) and feed them to the AI model for classification in real-time.

6. User Interface Development: Creating an intuitive user interface (UI) for interacting with the recycle bin system is part of the project scope. This includes designing and implementing a GUI that displays the live video feed, classification results, and any additional information or feedback for the user.

7. Integration and Deployment: Integrating all system components (including hardware, software, and user interface) and deploying the AI-powered recycle bin in a real-world environment is within the project scope. This involves ensuring compatibility, stability, and reliability of the system under real-world conditions.

8. Testing and Validation: Conducting thorough testing and validation of the system is crucial to ensure its functionality, accuracy, and robustness. The scope includes testing the system with various waste items, environmental conditions, and user interactions to identify and address any issues or limitations.

9. Maintenance and Updates: Once deployed, the project scope extends to ongoing maintenance, monitoring, and updates to ensure the continued performance and effectiveness of the AI video classifying recycle bin system. This may involve addressing issues, optimizing performance, and incorporating feedback from users or stakeholders.

10. Documentation and Knowledge Transfer: Documenting the project's design, implementation, and operation is essential for knowledge transfer and future reference. The scope includes preparing documentation such as user manuals, technical guides, and training materials to facilitate the use and maintenance of the system by stakeholders.

1.3 Need of the Project

The implementation of an AI video classifier for recycling bins addresses several pressing needs in waste management. Firstly, it offers a solution to the inefficiencies and inaccuracies inherent in traditional manual sorting methods, streamlining the recycling process by automating the identification and separation of recyclable and non-recyclable materials. This not only enhances operational efficiency but also mitigates the risk of contamination in recycling streams, improving the quality of recycled materials. Moreover, by promoting proper waste sorting practices, the AI-powered recycle bin encourages individuals to participate in recycling efforts, thereby fostering environmental sustainability and reducing the burden on landfills. Additionally, the project facilitates resource conservation by maximizing the recovery of valuable materials through accurate sorting, ultimately contributing to the conservation of natural resources. Beyond practical benefits, the project serves as a platform for education and awareness, providing users with insights

into recycling practices and environmental impact. Furthermore, the scalability and adaptability of AI-powered recycle bins enable their deployment in diverse settings, catering to the waste management needs of various sectors and communities. Overall, the project aligns with the goals of cost-effective waste management, regulatory compliance, and technological innovation, positioning it as a valuable solution to modern waste management challenges.

1. Efficient Waste Sorting: Traditional recycling methods often rely on manual sorting, which is time-consuming, labor-intensive, and prone to errors. An AI video classifying recycle bin offers an automated solution that can efficiently sort recyclable and non-recyclable items, thereby streamlining the recycling process.

2. Promotion of Recycling: Proper waste sorting is essential for effective recycling practices. By providing an intelligent recycle bin equipped with AI video classification capabilities, individuals are encouraged to participate in recycling efforts, contributing to environmental sustainability and waste reduction.

3. Reduction of Contamination: Contamination of recycling streams with non-recyclable materials poses a significant challenge for recycling facilities. An AI-powered recycle bin can accurately identify and separate contaminants, minimizing the risk of contamination and improving the quality of recycled materials.

4. Resource Conservation: Recycling conserves valuable natural resources by reducing the need for raw materials extraction and production. By enhancing the efficiency and accuracy of waste sorting through AI technology, a video classifying recycle bin helps maximize resource conservation and minimize environmental impact.

5. Cost Savings: Automating waste sorting with AI technology can lead to cost savings for municipalities, recycling facilities, and businesses involved in waste management. By reducing the need for manual labor and improving recycling efficiency, AI video classifying recycle bins can lower operational costs and increase overall productivity.

6. Education and Awareness: Incorporating AI technology into recycling bins provides an opportunity to educate individuals about proper waste sorting practices. Through interactive interfaces and feedback mechanisms, users can learn about recyclable materials, proper disposal methods, and the environmental benefits of recycling.

7. Scalability and Adaptability: AI-powered recycle bins can be deployed in various settings, including public spaces, commercial buildings, and residential areas. The scalability and adaptability of these systems make them suitable for addressing diverse waste management challenges in different contexts and environments.

8. Data-driven Insights: The use of AI technology enables the collection of data on waste generation patterns, recycling rates, and material compositions. Analyzing this data can provide valuable insights for optimizing waste management strategies, identifying opportunities for improvement, and informing policy decisions.

9. Compliance with Regulations: Many jurisdictions have regulations and mandates aimed at increasing recycling rates and reducing waste sent to landfills. Implementing AI video classifying recycle bins can help organizations and municipalities comply with these regulations by improving recycling efficiency and diverting more materials from landfills.

10. Innovation and Technological Advancement: Embracing AI technology in waste management represents a step towards innovation and technological advancement in the field. By harnessing the power of AI for waste sorting and recycling, we can leverage cutting-edge solutions to address environmental challenges and create a more sustainable future.

2. LITERATURE SURVEY

Literature Survey on AI Video Classifier for Recycling Bins:

1. "Deep Learning for Waste Management: A Survey" by Al-Ghannami et al. (2020):

This comprehensive survey provides an overview of deep learning techniques applied to various aspects of waste management, including waste classification and sorting. The paper discusses the use of convolutional neural networks (CNNs) for image and video-based waste classification tasks, highlighting their effectiveness in improving recycling efficiency.

2. "Automated Waste Sorting: A Survey of Technologies and Applications" by Pradhan et al. (2019):

This survey explores different technologies and methodologies for automated waste sorting, including AI-based approaches. It discusses the challenges and opportunities in waste sorting automation and reviews the latest advancements in AI techniques for waste classification, including video-based methods.

3. "Recent Advances in Waste Segregation Techniques: A Comprehensive Survey" by Gond et al. (2021):

This survey provides an in-depth analysis of recent advances in waste segregation techniques, with a focus on machine learning and AI-based approaches. It discusses the use of computer vision and deep learning algorithms for waste classification tasks and evaluates their performance in real-world applications.

4. "Deep Learning-Based Waste Sorting for Smart Cities: A Review" by Li et al. (2020):

This review paper discusses the role of deep learning techniques in enabling smart waste management systems for urban environments. It examines the challenges and opportunities in waste sorting automation and presents case studies and examples of AI-based waste sorting solutions, including video-based classification methods.

5. "Artificial Intelligence in Waste Management: Current Trends and Future Perspectives" by Patel et al. (2021):

This paper provides an overview of the current trends and future prospects of AI applications in waste management. It discusses the potential of AI technologies, including computer vision and machine learning, to revolutionize waste sorting and recycling processes, with a focus on video-based classification systems.

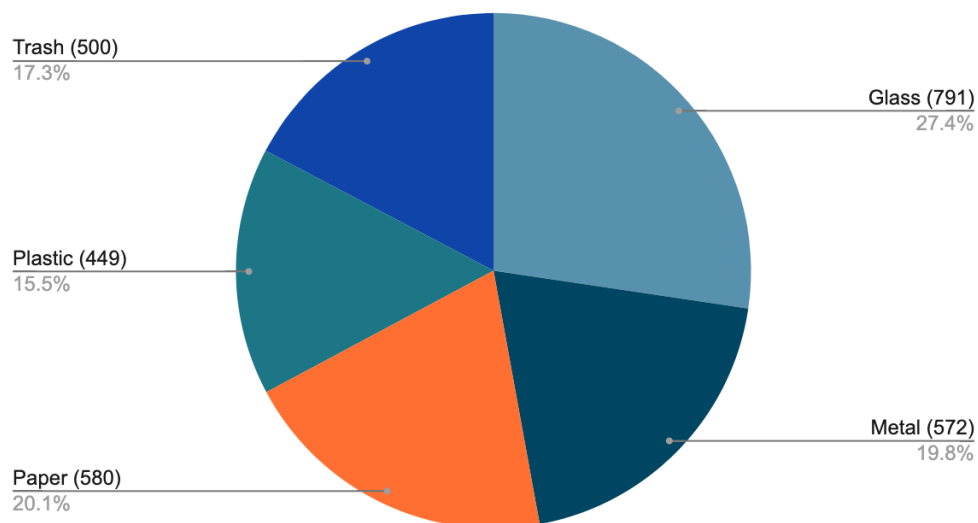
6. "Real-Time Waste Classification System Using Deep Learning" by Mohamed et al. (2019):

This research paper presents a real-time waste classification system based on deep learning techniques. It describes the implementation of a CNN-based model for waste classification using images and videos captured by a surveillance camera, demonstrating the feasibility and effectiveness of AI-based waste sorting solutions.

7. "A Review of Machine Learning Techniques for Automated Waste Sorting" by Siva et al. (2020):

This review paper provides an overview of machine learning techniques applied to automated waste sorting tasks. It discusses the advantages and limitations of different machine learning algorithms and explores their potential applications in waste classification, including video-based sorting systems. These literature sources collectively provide insights into the state-of-the-art techniques, challenges, and future directions of AI-based video classifier systems for recycling bins, offering valuable guidance for researchers and practitioners in the field.

Garbage Classification Dataset



3.PART LIST AND COST

3.1 Part list

Raspberry Pi 4 Model B	₹3400
Pi Camera	₹300
SD card	₹400
15W Adapter Raspberry Pi	₹725
PVC pipes	₹650
HDMI to USB C cable	₹600
Power supply 9V 1A	₹1000
Acrylic Sheet	₹540
PVC Foam Sheet	₹450
Castor Wheels and Magnets	₹100
Spray Paint	₹250
Jumper Wires	₹50
Servo MG 996 R	₹450
Miscellaneous Costs	₹3000(approx.)

Total cost- ₹12000

The entire cost sums up to ₹12k. This includes all the extra costs such as the extra items purchased, the faulty components and the unnecessary items that have been purchased in order for the completion of the project.

These extra costs have occurred to the fact that the project to be implemented is a prototype which requires a lot of R&D to execute and this project has built an experience factor in the trainees.

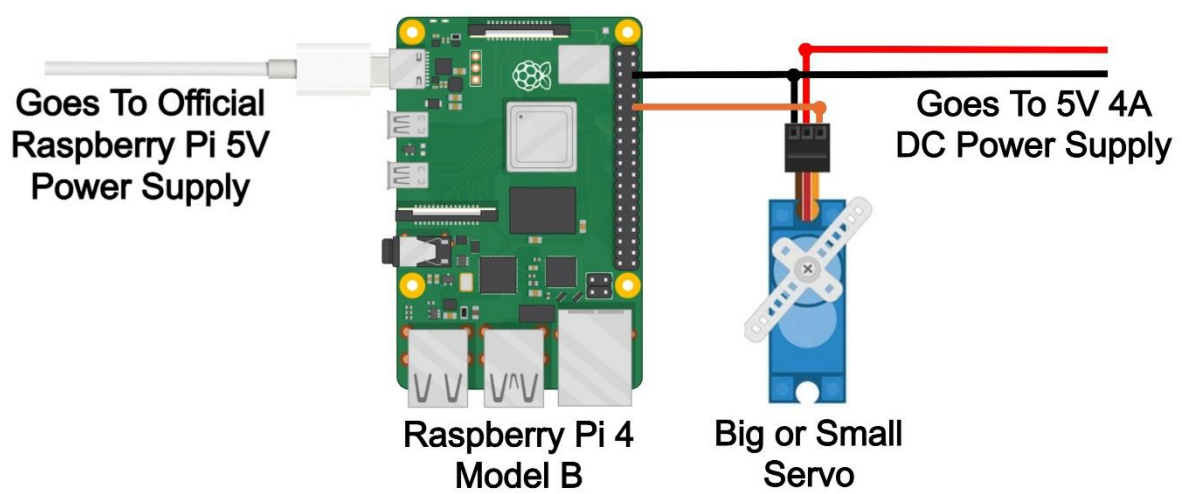
The technical knowledge and the skills cultivated during the project are extremely valuable and has helped us build a technical IQ to solve complex problems in the system.

The costing of the project has been calculated with respect to the entire expenditure occurred in the execution of the project. This includes the smallest of the items that were required to build the construction.

This costing also includes the travelling expenditure which was made to source out the components and the necessary items required to build the project and finish it with at most accuracy and perfection.

4.DESIGN AND IMPLEMENTAION

4.1 Circuit Diagram



4.2 Working Details

The recycle bin is equipped with a camera or multiple cameras to capture video footage of waste items being disposed of. The camera(s) continuously stream video input to the system for analysis. The incoming video frames undergo preprocessing steps to enhance their quality and prepare them for classification. Preprocessing may include resizing, normalization, and filtering to improve the efficiency and accuracy of the classification process.

The core of the system is an AI model, typically a convolutional neural network (CNN), trained to recognize and classify different types of waste items. The AI model is trained on a diverse dataset of labeled waste images or videos, learning to extract relevant features and patterns from the input data.

The AI model utilizes a classification algorithm to process each frame of the video footage and categorize the waste items into recyclable and non-recyclable classes. The algorithm applies the learned features from the training data to make predictions on unseen waste items in real-time.

The system performs real-time processing of the video stream, analyzing each frame as it is captured by the camera. This ensures timely classification of waste items as they are deposited into the recycle bin, enabling immediate feedback to users. Based on the classification results, the system makes decisions on how to handle each waste item. Recyclable items may be directed to the appropriate recycling container within the bin, while non-recyclable items may be diverted to a separate compartment or flagged for manual inspection.

The system may include a user interface (UI) to display the live video feed from the camera(s) along with the classification results. The UI provides real-time feedback to users, allowing them to monitor the recycling process and ensure proper disposal of waste items.

To enhance user engagement and awareness, the system may incorporate a feedback mechanism to provide users with information on their recycling behavior. This could include notifications, alerts, or rewards for proper waste sorting practices.

Once developed, the system is integrated into the recycle bin hardware and deployed in a real-world setting, such as a recycling facility, public space, or commercial building.

Continuous monitoring and maintenance ensure the proper functioning of the system over time. This model contains a raspberry pi which is capable of high AI image processing that is equipped with two servo motors which are responsible for the rotation of the lids and the compartments which is necessary for a smooth operation of the bin.

It also contains a frame that is constructed with the help of a PVC pipe which is very rigid, strong and provides for a high and stable performance.

4.3 How does the AI image detection work

AI image detection on a Raspberry Pi involves running a trained deep learning model, typically a convolutional neural network (CNN), on the Raspberry Pi's hardware to perform real-time object detection. Here's a general overview of how it works:

1. **Hardware Setup:** The Raspberry Pi is a small, low-cost, single-board computer that can be equipped with a camera module for capturing images. Additionally, it may be connected to other peripherals such as a display, keyboard, and mouse for user interaction.
2. **Software Installation:** To enable AI image detection on the Raspberry Pi, you need to install the necessary software libraries and frameworks. Common choices include TensorFlow Lite, OpenCV, and the Raspberry Pi's own operating system, Raspbian.
3. **Model Selection and Optimization:** Choose a pre-trained deep learning model suitable for object detection tasks, such as YOLO (You Only Look Once), SSD (Single Shot MultiBox Detector), or Mobile Net. These models are optimized for inference on resource-constrained devices like the Raspberry Pi and offer a good balance between accuracy and speed.
4. **Model Conversion:** Convert the selected deep learning model into a format compatible with the Raspberry Pi's hardware and software environment. This often involves converting the model to TensorFlow Lite format or another lightweight format optimized for edge devices.
5. **Inference:** Once the model is converted and loaded onto the Raspberry Pi, it can perform inference on input images captured by the camera module. The model analyzes each frame in real-time, detecting objects and drawing bounding boxes around them to indicate their location and class.
6. **Optimization for Speed:** To ensure real-time performance on the Raspberry Pi's limited hardware resources, optimize the inference process by utilizing techniques such as quantization, model pruning, and hardware acceleration (e.g., using the Raspberry Pi's GPU or specialized hardware like the Coral USB Accelerator).

7. Integration with Applications: Integrate the AI image detection functionality with applications running on the Raspberry Pi, such as home security systems, wildlife monitoring devices, or robotics projects. This involves designing user interfaces, handling input from sensors, and executing actions based on the detected objects.

8. Deployment: Deploy the Raspberry Pi-based AI image detection system in its target environment, ensuring proper setup and calibration. Monitor its performance and make adjustments as needed to optimize accuracy and reliability.

By leveraging the capabilities of the Raspberry Pi and deploying optimized deep learning models, AI image detection can be effectively implemented on edge devices for various practical applications, ranging from home automation to industrial monitoring.

Further even the pi cam needs to be connected to the raspberry pi 4 so that the image can be detected by the camera and the image can be sent to the raspberry pi for processing and the result can be produced by training the raspberry pi using google colab.

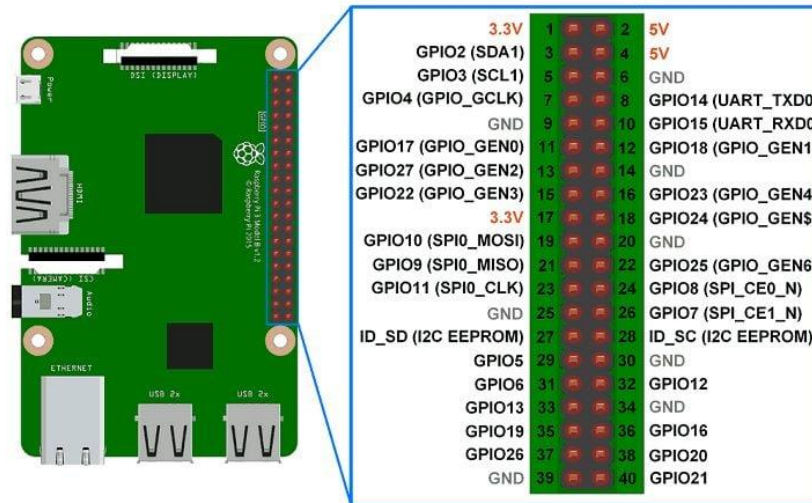
In order to connect the camera to the raspberry pi we must ensure that the raspberry pi is switched off so that we can avoid a system boot and damage. Now we can connect the camera to the port and start the system to enable the camera interface.

4.4 Technical Specifications

1. Raspberry pi 4 model B

The Raspberry Pi 4 Model B is the fourth generation of the popular single-board computer developed by the Raspberry Pi Foundation. It builds upon the success of its predecessors while introducing significant improvements in performance, connectivity, and versatility. Raspberry Pi 4 Model B is the latest product in the popular Raspberry Pi range of computers. It offers ground-breaking increases in processor speed, multimedia performance, memory, and connectivity compared to the prior-generation Raspberry Pi 3 Model B+, while retaining backwards compatibility and similar power consumption. For the end user, Raspberry Pi 4 Model B provides desktop performance comparable to entry-level x86 PC systems. This product's key features include a high-performance 64-bit quad-core processor, dual-display support at resolutions up to 4K via a pair of micro-HDMI ports, hardware video decode at up to 4Kp60, up to 4GB of RAM, dual-band 2.4/5.0 GHz wireless

LAN, Bluetooth 5.0, Gigabit Ethernet, USB 3.0, and PoE capability (via a separate PoE HAT add-on). The dual-band wireless LAN and Bluetooth have modular compliance certification, allowing the board to be designed into end products with significantly reduced compliance testing, improving both cost and time to market.



2. Castor wheels and magnets

A caster (or castor) is an undriven wheel that is designed to be attached to the bottom of a larger object (the "vehicle") to enable that object to be moved. A caster is a non-powered wheel that is designed to be fitted to the bottom of a bigger object and used to move it. Caster wheels are used in shopping carts, office chairs, medical beds, and material handling equipment. Magnets here can be used to stick the garbage compartment in place so that it does not move while operation and also to hold it in place.



3. Pi Camera

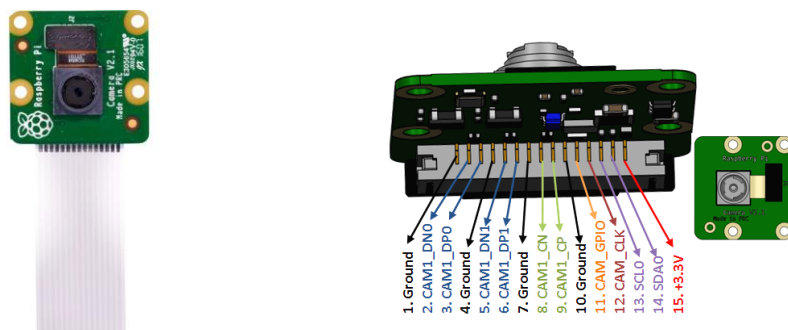
This 5-megapixel sensor with OV5647 camera module is capable of 1080p video and still images that connect directly to your Raspberry Pi. This is the plug-and-play-compatible latest version of the Raspbian operating system, making it perfect for time-lapse photography, recording video, motion detection and security applications. Connect the included ribbon cable to the CSI (Camera Serial Interface) port on your Raspberry Pi, and you are good to go!

The board itself is tiny, at around 25mm x 23mm x 9mm and weighing in at just over 3g, making it perfect for mobile or other applications where size and weight are important. The sensor has a native resolution of 5 megapixel, and has a fixed focus lens on board. In terms of still images, the camera is capable of 2592 x 1944-pixel static images, and also supports 1080p30, 720p60 and 640x480p90 video.

Note: This module is only capable of taking pictures and video, not sound.

Features

- Compatible with Raspberry Pi 4 Model B/3B+/3B/2B/Zero Wireless
- 5 Megapixel OV5647 Camera
- Camera specifications
 - Static Images Resolution: 2592×1944
 - Supported Video Resolution: 1080p/30 fps, 720p/ 60fps and 640 x480p 60/90 video recording
 - Aperture (F): 1.8
 - Visual Angle: 65 degree
- Dimension: 24mmx23.5mmx8mm
- Weight: 3g
- Interface: CSI connector
- Supported OS: Raspbian (latest version recommended)



4. SD Card

An SD (Secure Digital) card is a type of removable flash memory storage device commonly used in portable electronic devices such as digital cameras, smartphones, tablets, and single-board computers like the Raspberry Pi. Here's a description of an SD card:

Form Factor: SD cards are small, rectangular-shaped devices with dimensions standardized by the Secure Digital Association. They come in various sizes, including standard SD, miniSD, and microSD, with microSD being the most common form factor used in modern devices due to its compact size.

Storage Capacity: SD cards are available in a range of storage capacities, from a few gigabytes (GB) to multiple terabytes (TB). The storage capacity determines how much data the card can hold, including photos, videos, music, documents, and software applications.

Storage Medium: SD cards use NAND flash memory technology to store data. NAND flash memory consists of memory cells organized into blocks and pages, allowing for high-speed read and write operations. The memory cells retain data even when the power is turned off, making SD cards a non-volatile storage medium.

Speed Class: SD cards are classified based on their speed performance, which determines how quickly data can be read from or written to the card. Speed classes include Class 2, 4, 6, 10, and UHS (Ultra High Speed) Speed Classes such as U1 and U3. The higher the speed class, the faster the card's data transfer rates.

File System: SD cards typically use the FAT32 or exFAT file systems, which are compatible with a wide range of operating systems and devices. The file system defines how data is organized and accessed on the card, facilitating data storage and retrieval.



5. 15W Adapter Raspberry Pi

The official Raspberry Pi USB-C Power Supply is designed to power Raspberry Pi 4 Model B and Raspberry Pi 400 computers. Featuring a captive USB-C cable, the power supply is available in five different models to suit different international power sockets, and in two colors, white and black.

The adapter comes with 1.5m 18 AWG captive cable with USB-C output connector. The design of the adapter with a Raspberry LOGO engraved on the top which makes it different from any other adapter available in the market.

The Raspberry Pi 15W USB-C Power Supply is designed to power Raspberry Pi 4 and Raspberry Pi 400 computers. Featuring a 1.5m captive USB cable, this reliable, high-quality power supply provides an output of 5.1V / 3.0A DC via a USB-C connector, for all the power your Raspberry Pi 4 or 400 will need.

The Raspberry Pi 15W USB-C Power Supply is designed to power Raspberry Pi 4 and Raspberry Pi 400 computers. Featuring a 1.5m captive USB cable, this reliable, high-quality power supply provides an output of 5.1V / 3.0A DC via a USB-C connector, for all the power your Raspberry Pi 4 or 400 will need.



6. PVC Pipes

The third-most widely used synthetic polymer, PVC, is manufactured and used at the rate of 40 million tons a year worldwide. Its main application is as a conduit for water in various contexts.

What is made of, which makes it a universally sought-after product? The raw material used in it is polyvinyl chloride, whose multitude of beneficial physical and chemical properties make it the ideal choice for a safe and cost-effective channel in a spectrum of infrastructure applications, the foremost being water supply.

PVC stands for polyvinyl chloride. It is a chlorinated hydrocarbon polymer. In its natural state, it is rigid and brittle. But when combined with additives such as plasticizers, it becomes more resilient and malleable.

Some of its applications are in electrical insulation, medical tubing, flooring, furniture, signage and as a substitute for rubber. But its most widespread use is in the manufacture of pipes, which are used in water supply, plumbing and irrigation.

Depending on the application, a few variants of PVC pipes are used as hot or cold-water conduits in industrial as well as commercial contexts. PVC pipes have been used for over 80 years for various commercial purposes. In the United States alone, two million miles of PVC pipelines are functional today.

PVC pipes are ubiquitous in the context of water supply. The pipelines used for water mains almost entirely consist of PVC pipes. Sanitation sewer networks and large-scale irrigation networks also comprise PVC pipes.

In the residential and commercial domain, PVC pipes are used in plumbing, drainage, , agricultural irrigation, electricity & telecommunications systems and the ductwork of heating and cooling systems.

Underground PVC pipes have a larger bore and a thicker wall and are used by utility companies to supply drinking water through a network to residential and commercial buildings.



7. HDMI to USB C Cable

Raspberry Pi micro-HDMI to HDMI cable (male-male) is the one you'll want for your Raspberry Pi 4. It's sturdy, strong, and even has a cute little Raspberry Pi logo on the connectors!

This cable will connect your Pi 4 with any External devices which support up to a 4K resolution and high-speed data transmission. Watch the video, stream media, with multiple audio formats from stereo to multi-channel sound and view documents through your TV Screen.

This RoHS-compliant cable allows you to transfer movies, photos, and videos from devices like smartphones, digital cameras, and video cameras with micro-HDMI connectors to view on your standard HDMI (type-A) television, projector, or computer monitor.

This is a must-have accessory for those who use the latest gadgets and electronics developments boards like Raspberry Pi 4 as well as who would like to watch videos and work on the big screen.

- Fully compatible with High-Speed HDMI with Ethernet specification (Version 1.4).
- A superb cable with excellent audio-visual transfer properties that deliver signal without compromising purity and balance.
- Nickel-plated connectors for highest signal transfer rate and resistance in corrosion.
- Large gauge PVC jacket provides maximum shielding against wear and tear and extreme temperatures.
- Reinforced quad-layer braided shielding and Mylar-foil shielding provides utmost protection against RF and EM interference.
- Moulded strain-relief design lessens plug/socket pressure.
- Designed to meet all HDMI standards.

- Supports 480i, 480p, 720p, 1080i, 1080p, 1440p and 4K@60p resolution HDMI plug to HDMI plug leads
- Compatible with Raspberry Pi 4 Model B and all previous Raspberry Pi models



8. Power Supply 9V 1A

A 9V 1A power supply is an electrical device designed to provide a stable and regulated voltage of 9 volts (9V) and a maximum current output of 1 ampere (1A). Here's a description of a 9V 1A power supply:

1. **Voltage Output:** The power supply delivers a constant voltage output of 9 volts (9V). This voltage level is suitable for powering a wide range of electronic devices, including small appliances, electronic gadgets, and low-power devices such as LED lights, sensors, and microcontrollers.
2. **Current Output:** The power supply is capable of supplying a maximum current output of 1 ampere (1A). This current rating determines the maximum amount of electrical current that the power supply can deliver to the connected device without exceeding its specified limits.
3. **Regulated Output:** A 9V 1A power supply typically features regulated output voltage, ensuring a stable and consistent voltage level regardless of fluctuations in input voltage or load conditions. This regulation helps protect connected devices from damage due to overvoltage or voltage spikes.
4. **Connector Type:** The power supply may come with various types of connectors, such as barrel connectors, coaxial connectors, or USB connectors, depending on the specific application and compatibility requirements. It is essential to ensure that the connector type matches the input requirements of the device being powered.

5. **Input Voltage:** The power supply operates from a standard AC (alternating current) input voltage, typically ranging from 100 volts to 240 volts AC. This wide input voltage range allows the power supply to be used in different regions with varying mains power standards.

6. **Safety Features:** A 9V 1A power supply may include built-in safety features such as overcurrent protection, overvoltage protection, and short-circuit protection to safeguard both the power supply and the connected devices from potential damage caused by electrical faults or malfunctions.

7.***Compact and Lightweight Design:** Many 9V 1A power supplies are designed to be compact, lightweight, and portable, making them convenient for use in various settings, including home, office, or travel. Their small size and low weight make them easy to carry and store when not in use.

8. **Versatile Applications:** A 9V 1A power supply is suitable for powering a wide range of electronic devices, including routers, modems, audio equipment, musical instruments, toys, and hobbyist projects. It is commonly used in applications where a stable 9V power source is required to operate or charge electronic devices efficiently.

Overall, a 9V 1A power supply provides a reliable and regulated power source for powering various electronic devices and applications, offering convenience, safety, and versatility in a compact and affordable package.



9. Acrylic Sheet

Acrylic, also known as Plexiglass, is a versatile plastic material with a variety of purposes and benefits, available in a spectrum of colors and opacities. Acrylic plastic was first produced in 1928 and brought to the market by Rohm and Hass Company around 1933. It was initially used during World War II for products such as airplane windows, canopies, and turrets.

Today, acrylic is most commonly sold in sheets of various thicknesses but can also be found in forms such as rods or tubes, and styles such as frosted, mirrored, or non-glare. Not only is acrylic available in many shapes and sizes, they can also be molded and colored to fit specific applications. No matter what option best suits your needs, acrylic will provide a cost-effective, durable material for your project. Clear acrylic plexiglass

Since acrylic combines a wide variety of options with a copious amount of benefits, they can be fashioned into a diverse range of useful products. This includes aquariums, retail displays, window panes, and even bulletproof glass! Acrylic is almost certainly bound to be an exemplary choice no matter what purpose it is used for.

Acrylic plastic is a transparent thermoplastic homopolymer that is often colloquially referred to as plexiglass. This plastic has unique properties that make it ideal for a variety of purposes, ranging from basic household items to fiber optic cables that power the world. These properties have allowed acrylic to enjoy immense popularity in manufacturing as well as DIY projects:

- High impact resistance
- High optical clarity
- Innate weatherability and UV resistance
- Excellent dimensional stability
- Lightweight
- Excellent chemical resistance



10.PVC Foam Sheet

PVC foam, also known as polyvinyl chloride foam, is a durable, closed cell, free foam PVC sheet material. This material is easy to work with, and can be easily sawed, die-cut, drilled, or bounded to meet the needs of numerous applications. PVC foam offers many advantages, such as superior impact resistance, high strength, great durability, low water absorption, high corrosion resistance, and fire resistance to name a few. These benefits provide the perfect material to withstand all types of indoor and outdoor conditions, including harsh

weather. PVC foam is also a great alternative to other materials, such as wood or aluminum, and can often last 40 years without experiencing any damage. The price range of this material will vary compared to these traditional materials, but PVC Foam does not require the extra maintenance to keep the material in optimal condition.

Signage made from PVC foam offers great value for a variety of sign applications. PVC foam sheet is durable and displays colors consistently, making it the perfect choice for directional signage, POS displays, display boards, menu boards, and real estate signs.

The Building and Construction market has also seen tremendous improvements with PVC foam applications. Architectural columns, decorative shutters, store fixtures, shower pans, furniture, outdoor kitchens, and other anti-microbial solutions are just a few examples; but the versatility of PVC foam allows builders, architects, and designers to get as creative as they want to with their designs.

Exhibit spaces have also benefited from PVC foam applications. Because PVC foam is lightweight and easy to work with, trade show booths, display cases, staging and event displays, and kiosks made from this material have become easier to install and use.

For marine market applications, marine grade PVC foam has been specifically designed to withstand harsh environments. Marine upholstery substrate, swim platforms, coring material for backing plates, and cup holders can all be made from this marine PVC foam and have seen excellent performance.



11.Spray Paint

Spray paint (formally aerosol paint) is paint that comes in a sealed, pressurized container and is released in an aerosol spray when a valve button is depressed. The propellant is what the container of pressurized gas is called. When the pressure holding the gas is released through the valve, the aerosol paint releases as a fine spray.[1] Aerosol painting is one form of spray painting; it leaves a smooth, even coat, unlike many traditional rolled and brushed paints. Aerosol primer can be applied directly to bare metal and many plastics.

Most aerosol paints also have a metal, marble, glass or plastic ball called a pea inside of the can, which is used to mix the paint when the can is shaken.

Aerosol propellants consist of flammable gas combinations that can lead to fire and explosions if they are sprayed, punctured, or exposed to flames.

Aerosol paint needs different types of plastic-like polymers to make it work. The most commonly used are alkyds and acrylics. Polyvinyl acetate (PVA) is another substitute for acrylics. Different companies will use specific mixes of polymers and plasticizers (like dibutyl phthalate and dibutyl maleate) to make the spray paint flexible and durable.

Most brands include a wide variety of paints, including primers, heat, and traffic resistant enamels, gloss and matte finishes, metallic colors, and textured paints for home decor.

Aerosol paint is useful for quick, semi-permanent marking on construction and surveying sites. Inverted cans for street, utility or field marking can be used upside-down with an extension pole. APWA (American Public Works Association) has standardized colors for utility and excavation markings. Hiking trails can also be marked with aerosol paint trail blazes.

Small to medium-sized repairs to automobile bodywork can be completed by enthusiasts at home using aerosol paint, though to paint an entire vehicle in this manner would be difficult and expensive. The main disadvantages, compared to a professional spray gun, include the limited quality provided by the built-in nozzle and the lack of infrared baking after applying the paint, which indicates that the paint could take several months to obtain its final hardness.



12.Jumper Wires

Jumper wires are simply wiring that have connector pins at each end, allowing them to be used to connect two points to each other without soldering. Jumper wires are typically used with breadboards and other prototyping tools in order to make it easy to change a circuit as needed. Fairly simple. In fact, it doesn't get much more basic than jumper wires.

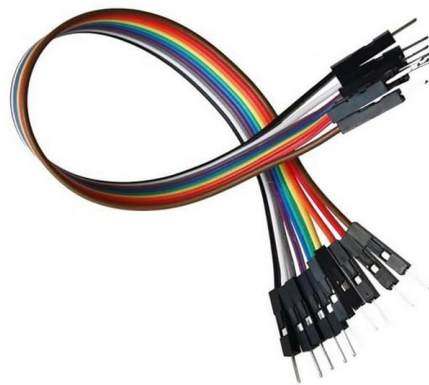
Though jumper wires come in a variety of colors, the colors don't actually mean anything. This means that a red jumper wire is technically the same as a black one. But the colors can be used to your advantage in order to differentiate between types of connections, such as ground or power.

While jumper wires are easy and inexpensive to purchase, it can also be a fun task to challenge students to make their own. Doing so requires insulated wire and wire strippers. However, beware that it is important not to nick the wire when stripping off the insulation

Jumper wires typically come in three versions: male-to-male, male-to-female and female-to-female. The difference between each is in the end point of the wire. Male ends have a pin protruding and can plug into things, while female ends do not and are used to plug things into. Male-to-male jumper wires are the most common and what you likely will use most often. When connecting two ports on a breadboard, a male-to-male wire is what you'll need.

A jumper wire is a short-insulated wire with bare (stripped of insulation) ends. You use jumper wires, such as the one shown in the following figure, to connect two points in a breadboard circuit. Even if you have a set of precut jumper wires, chances are you'll need to make a jumper wire of a specific length for a circuit or two.

Alligator clips, which consist of two spring metal clips connected by wire, are actually just fancy jumper wires! Their unique connection point (they can be clipped on instead of inserted into) allow alligator clips to be used in a variety of situations that would get a little awkward with a traditional jumper wire. One of the most common uses for alligator clips in education is with the Makers, though they can also be used to connect the ports on a Lily Pad board as well as for a number of other applications



13.Servo MG 996 R

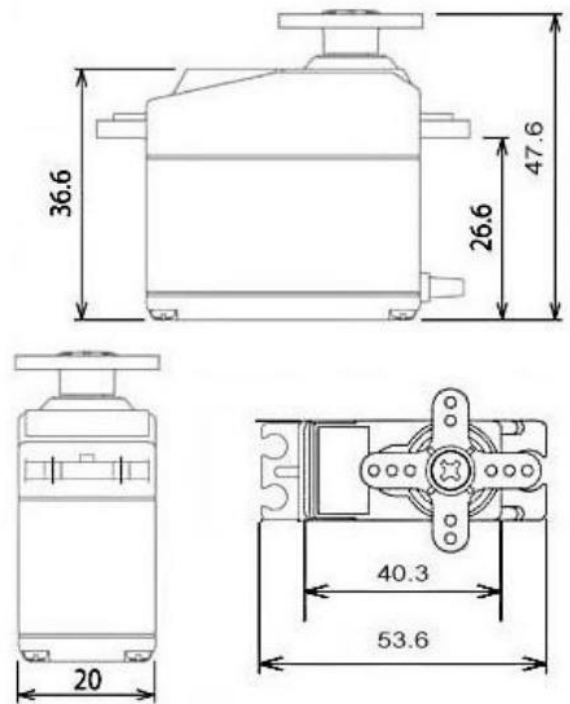
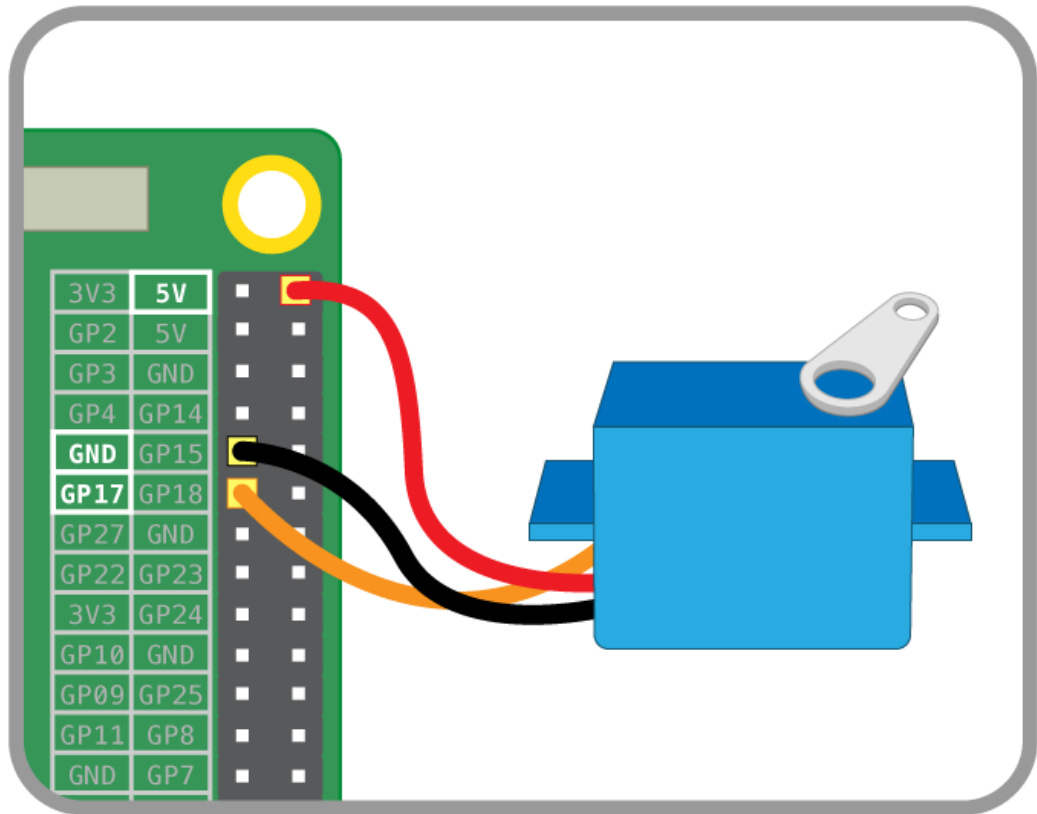
This High-Torque MG996R Digital Servo features metal gearing resulting in extra high 10kg stalling torque in a tiny package. The MG996R is essentially an upgraded version of the famous MG995 servo, and features upgraded shock-proofing and a redesigned PCB and IC control system that make it much more accurate than its predecessor.

The gearing and motor have also been upgraded to improve dead bandwidth and centering. The unit comes complete with 30cm wire and 3 pin 'S' type female header connector that fits most receivers, including Futaba, JR, GWS, Cirrus, Blue Bird, Blue Arrow, Corona, Berg, Spektrum and Hitech.

This high-torque standard servo can rotate approximately 120 degrees (60 in each direction). You can use any servo code, hardware or library to control these servos, so it's great for beginners who want to make stuff move without building a motor controller with feedback & gear box, especially since it will fit in small places.

The MG996R Metal Gear Servo also comes with a selection of arms and hardware to get you set up nice and fast

- Weight: 55 g
 - Dimension: 40.7 x 19.7 x 42.9 mm approx.
 - Stall torque: 9.4 kgf·cm (4.8 V), 11 kgf·cm (6 V)
 - Operating speed: 0.17 s/60° (4.8 V), 0.14 s/60° (6 V)
 - Operating voltage: 4.8 V a 7.2 V
 - Running Current 500 mA –
 - Stall Current 2.5 A (6V)
 - Dead band width: 5 µs
 - Stable and shock proof double ball bearing design
 - Temperature range: 0 °C –
- 4.8 V a 7.2 V
- 900 mA (6V)
- double ball bearing design



The above mentioned components are the most essential components that have been used in the construction of the project and play a significant role in the working of the project.

These components have been tuned using the trail and error method and have been calibrated accordingly in order to function according to our needs.

The raspberry pi being used is the controller that controls the entire process and is responsible for the implementation of AI in the project. This machine is a very capable one with high power and speed to process the ai stuff.

The technical knowledge and the skills cultivated during the project are extremely valuable and has helped us build a technical IQ to solve complex problems in the system.

4.5 Features of the technical specifications

1. Camera:

- High-resolution camera capable of capturing clear video footage.
- Wide field of view to cover the entire area of the recycle bin.
- Low-light performance for operation in various lighting conditions.

2. Processing Unit:

- Powerful processor capable of real-time video analysis and classification.
- Sufficient memory (RAM) and storage (e.g., microSD card) for storing and processing video data.
- Support for hardware acceleration to improve performance, such as GPU or dedicated AI accelerators.

3. AI Model:

- Convolutional Neural Network (CNN) trained for video classification of recyclable and non-recyclable items.
- Optimized for efficiency and accuracy in real-time inference on resource-constrained hardware.
- Ability to handle diverse types of waste items and environmental conditions.

4. Classification Algorithm:

- Robust algorithm for detecting and classifying objects within video frames.
- Support for multi-class classification (e.g., recyclable, non-recyclable, organic).
- Integration of post-processing techniques for refining classification results and reducing false positives.

5. User Interface:

- Graphical User Interface (GUI) for displaying live video feed and classification results.
- Intuitive design with clear visual indicators for recyclable and non-recyclable items.
- Interactive features for user feedback and engagement (e.g., notifications, progress tracking).

6. Connectivity:

- Wi-Fi and/or Ethernet connectivity for remote monitoring and management.
- USB ports for connecting external devices, such as storage drives or peripherals.
- Compatibility with standard communication protocols for data exchange with other systems (e.g., MQTT for IoT integration).

7. Power Supply:

- Stable power source (e.g., 9V DC) with sufficient capacity to power the entire system.
- Energy-efficient design to minimize power consumption and extend battery life (if applicable).

8. Enclosure:

- Durable and weather-resistant enclosure suitable for outdoor or indoor deployment.
- Adequate ventilation to prevent overheating of internal components.
- Mounting options for easy installation and secure attachment to the recycle bin or surrounding infrastructure.

9. Integration and Deployment:

- Compatibility with existing waste management systems and infrastructure.
- Ease of installation and configuration, with clear documentation and setup instructions.

- Support for remote updates and maintenance to ensure continuous operation and performance optimization.

10. Compliance and Regulations:

- Compliance with relevant industry standards and regulations for waste management and environmental protection.
- Consideration of privacy regulations regarding the collection and processing of video data.

These technical specifications outline the key requirements for designing and implementing an AI video classifier recycle bin, ensuring efficient and accurate waste sorting while addressing practical considerations such as usability, connectivity, and compliance.

5.PROGRAM DEVELOPMENT

```
import argparse
import sys
import time
import RPi.GPIO as GPIO

import cv2
import mediapipe as mp

from mediapipe.tasks import python
from mediapipe.tasks.python import vision

from utils import visualize
from picamera2 import Picamera2
import RPi.GPIO as GPIO
import time

def setAngle1(angle):
    duty = angle / 18 + 3
    GPIO.output(18, True)
    p.ChangeDutyCycle(duty)
    time.sleep(1)
    GPIO.output(18, False)
    p.ChangeDutyCycle(duty)

def setAngle2(angle):
    duty = angle / 18 + 3
    GPIO.output(19, True)
    p.ChangeDutyCycle(duty)
```



```

time.sleep(1)
GPIO.output(19, False)
p.ChangeDutyCycle(duty)

GPIO.setmode(GPIO.BCM)
GPIO.setup(18, GPIO.OUT)
GPIO.setup(19, GPIO.OUT)

global x
x=0

p = GPIO.PWM(18, 50) # PWM frequency is 50Hz
p.start(2.5) # Initialization

# Global variables to calculate FPS
COUNTER, FPS = 0, 0
START_TIME = time.time()
picam2 = Picamera2()
picam2.preview_configuration.main.size = (640,480)
picam2.preview_configuration.main.format = "RGB888"
picam2.preview_configuration.align()
picam2.configure("preview")
picam2.start()

def run(model: str, max_results: int, score_threshold: float,
        camera_id: int, width: int, height: int) -> None:
    """Continuously run inference on images acquired from the camera.

```

Args:

model: Name of the TFLite object detection model.

max_results: Max number of detection results.

score_threshold: The score threshold of detection results.

camera_id: The camera id to be passed to OpenCV.

width: The width of the frame captured from the camera.

height: The height of the frame captured from the camera.

"""

Visualization parameters

row_size = 50 # pixels

left_margin = 24 # pixels

text_color = (0, 0, 0) # black

font_size = 1

font_thickness = 1

fps_avg_frame_count = 10

detection_frame = None

detection_result_list = []

def save_result(result: vision.ObjectDetectorResult, unused_output_image: mp.Image,
timestamp_ms: int):

global FPS, COUNTER, START_TIME

Calculate the FPS

if COUNTER % fps_avg_frame_count == 0:

FPS = fps_avg_frame_count / (time.time() - START_TIME)

START_TIME = time.time()

```

for detection in result.detections:

    for category in detection.categories:

        category_name = category.category_name

        # Do something with the category name, such as appending it to a list
global x
if category_name=="metal" and (x==0 or x==2 or x==3):

    print("Detected category1:", category_name)

    setAngle1(0)
    x=1
    time.sleep(3)
    setAngle2(180)
    time.sleep(3)
    setAngle2(0)

elif category_name=="paper" and (x==0 or x==1 or x==3):

    print("Detected category:2", category_name)

    setAngle1(70)
    x=2
    time.sleep(3)
    setAngle2(180)
    time.sleep(3)
    setAngle2(0)
    time.sleep(1)
    setAngle1(0)

elif category_name=="plastic" and (x==0 or x==2 or x==1):

    print("Detected category:3", category_name)

```

```
setAngle1(180)
x=3
time.sleep(3)
setAngle2(180)
time.sleep(3)
setAngle2(0)
time.sleep(1)
setAngle1(0)
```

```
detection_result_list.append(result)
COUNTER += 1
```

```
# Initialize the object detection model
base_options = python.BaseOptions(model_asset_path=model)
#time.sleep(3)
options = vision.ObjectDetectorOptions(base_options=base_options,
                                       running_mode=vision.RunningMode.LIVE_STREAM,
                                       max_results=max_results, score_threshold=score_threshold,
                                       result_callback=save_result)
detector = vision.ObjectDetector.create_from_options(options)

# Continuously capture images from the camera and run inference
while True:
    im= picam2.capture_array()
    # success, image = cap.read()
```

```

image=cv2.resize(im,(640,480))
image = cv2.flip(image, -1)

# Convert the image from BGR to RGB as required by the TFLite model.
rgb_image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
mp_image = mp.Image(image_format=mp.ImageFormat.SRGB, data=rgb_image)

# Run object detection using the model.
detector.detect_async(mp_image, time.time_ns() // 1_000_000)

# Show the FPS
fps_text = 'FPS = {:.1f}'.format(FPS)
text_location = (left_margin, row_size)
current_frame = image
cv2.putText(current_frame, fps_text, text_location, cv2.FONT_HERSHEY_DUPLEX,
            font_size, text_color, font_thickness, cv2.LINE_AA)

if detection_result_list:
    #print(detection_result_list[0])

    current_frame = visualize(current_frame, detection_result_list[0])
    detection_frame = current_frame
    detection_result_list.clear()

if detection_frame is not None:
    cv2.imshow('object_detection', detection_frame)

# Stop the program if the ESC key is pressed.
if cv2.waitKey(1) == 27:
    break

```

```

detector.close()

cap.release()

cv2.destroyAllWindows()


def main():
    parser = argparse.ArgumentParser(
        formatter_class=argparse.ArgumentDefaultsHelpFormatter)
    parser.add_argument(
        '--model',
        help='Path of the object detection model.',
        required=False,
        # default='efficientdet_lite0.tflite')
        default='best.tflite')
    parser.add_argument(
        '--maxResults',
        help='Max number of detection results.',
        required=False,
        default=5)
    parser.add_argument(
        '--scoreThreshold',
        help='The score threshold of detection results.',
        required=False,
        type=float,
        default=0.25)

    # Finding the camera ID can be very reliant on platform-dependent methods.

    # One common approach is to use the fact that camera IDs are usually indexed
    sequentially by the OS, starting from 0.

    # Here, we use OpenCV and create a VideoCapture object for each potential ID with 'cap
    = cv2.VideoCapture(i)'.

```

```

# If 'cap' is None or not 'cap.isOpened()', it indicates the camera ID is not available.
parser.add_argument(
    '--cameraId', help='Id of camera.', required=False, type=int, default=0)
parser.add_argument(
    '--frameWidth',
    help='Width of frame to capture from camera.',
    required=False,
    type=int,
    default=640)
parser.add_argument(
    '--frameHeight',
    help='Height of frame to capture from camera.',
    required=False,
    type=int,
    default=480)
args = parser.parse_args()

run(args.model, int(args.maxResults),
    args.scoreThreshold, int(args.cameraId), args.frameWidth, args.frameHeight)

if __name__ == '__main__':
    main()

```

5.1 explanation of the code

This Python script is designed to run an AI-based object detection model on a Raspberry Pi with a connected camera (presumably the PiCamera). Here's an explanation of the code:

1.Imports: The script imports necessary libraries and modules such as argparse for parsing command-line arguments, sys for system-specific functions and parameters, time for time-related functions, RPi.GPIO for controlling GPIO pins on the Raspberry Pi, OpenCV (cv2) for image processing, Mediapipe for AI-based object detection, and other custom modules.

2.GPIO Setup: The script sets up GPIO pins on the Raspberry Pi for controlling servos. Functions setAngle1() and setAngle2() are defined to control the angles of two servos connected to GPIO pins 18 and 19, respectively.

3.Camera Setup: The script initializes the PiCamera and configures its preview settings.

4.Main Function: The run() function is defined to continuously capture images from the camera, run inference using an object detection model (specified by the model argument), and visualize the detection results. The save_result() function handles the detection results and controls the servos based on the detected object categories.

5.Argument Parsing: The main() function parses command-line arguments using argparse. Arguments include the path to the object detection model (model), maximum number of detection results (maxResults), score threshold for detection results (scoreThreshold), camera ID (cameraId), frame width (frameWidth), and frame height (frameHeight).

6.Execution: Finally, the main() function is called if the script is executed directly (if __name__ == '__main__':). It parses the command-line arguments and runs the run() function with the specified arguments.

Overall, this script integrates object detection with servo control to create an AI video classifier recycle bin system on a Raspberry Pi. It continuously monitors the camera feed, detects objects using a pre-trained model, and triggers actions (e.g., moving the servos) based on the detected object categories (e.g., metal, paper, plastic).

6.HARDWARE DESIGN

The hardware design of the project consists of the following equipment that have made the project possible and also give a definition to the project

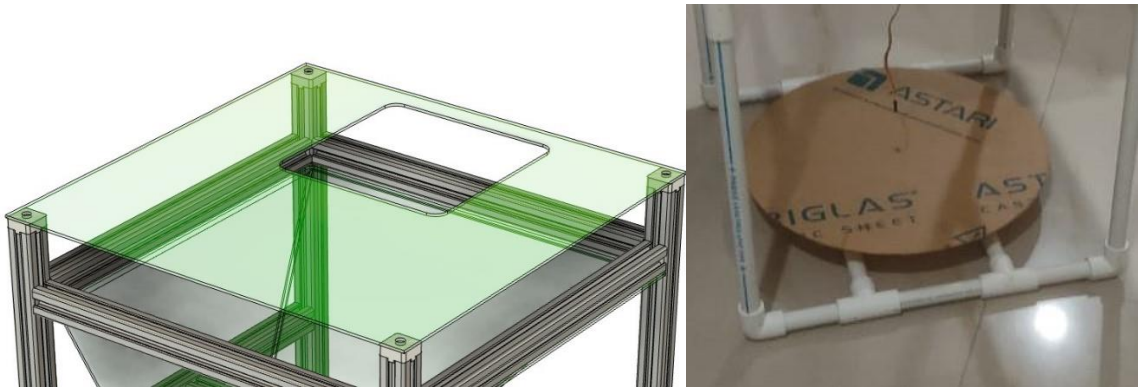
- **PVC Pipes**

The PVC pipes can be used for making the frame of the project. This constructed frame will be very rigid and will provide good support to the entire system and also due to the usage of the PVC pipes. The PVC pipes have been cut in a way to accommodate various kinds of joints to joint it in such a way that the structure resembles to a garbage bin that can be used for disposing the waste. The PVC was chosen in this project due to its low cost and good properties that help in the construction of a rigid frame. The PVC pipes house the entire mechanism and support it during the working stage.



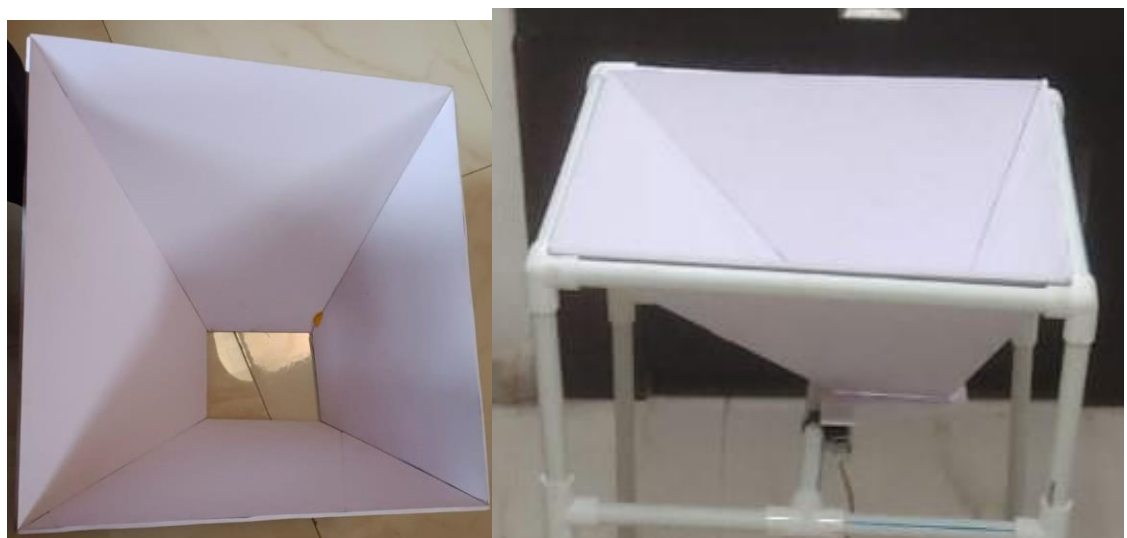
- **Acrylic Sheet**

The acrylic sheet has been used here in order to make the cover of the entire bin. The acrylic sheet is very strong and rigid and it is also transparent which allows us to see through the sheet. The lid that is being constructed has a slot inside it so that the waste can be accurately thrown in the bin and allows us to see what kind of waste is being thrown in the bin to verify if the model is working properly or not. The acrylic sheet is also used to make a circular plate which supports the containers.



- **PVC Foam Sheet**

The PVC foam sheet is used here in order to make the major part of the construction hence it plays a very important role in the project. It is used to construct an inverted prism shaped structure that guides the waste to fall on the opening of the bin which is also constructed with the help of the foam sheet. The containers in which the waste will be thrown is also constructed with the help of foam sheet. The foam sheet is inexpensive and provides good flexibility and strength in the construction.



- **Raspberry Pi 4 model B**

The raspberry pi is the boss of the entire project due to the fact that it is responsible for the entire controlling process and also it is the one that integrates the servos and the AI to execute the program. The raspberry pi has the object detection program in it which was completed by google colab so that when ever the camera and the image is detected the pi4 can classify it. It is also responsible for controlling the servo motors and the power supply is given to them separately. It is trained to identify the images and give the desired output.



- **Servo motor**

The servo motor is a motor that can be used to control the opening and the closing of the lid and the containers when the program is active and running. The servo rotates the opening and the waste is dropped into the containers. When the image is detected and identified first the container's position is adjusted and then the opening is opened to drop the waste in the container and after a delay the containers will return to their normal position and the opening will close.



- **Pi Camera**

The pi camera is a device that is responsible for detecting the image in the surroundings and then sending the signal to the raspberry for the identification of the image. The Pi camera is the most important part of the circuit because it captures the video and then send the data to receive an output. The pi cam operates on a low fps and has a 5mp camera for the video.



- **Power Supply**

The power supply is supposed to give the energy to the electronic components operate in a free manner without any lag. The power supply is connected to the raspberry pi to power it and also a separate power supply is connected to the servos to regulate the flow. The power supply being used here is 9V 1A power supply.



- **Understanding the Hardware and the circuit connections**

The PVC pipes can be used for making the frame of the project. This constructed frame will be very rigid and will provide good support to the entire system and also due to the usage of the PVC pipes. The PVC pipes have been cut in a way to accommodate various kinds of joints to joint it in such a way that the structure resembles to a garbage bin that can be used for disposing the waste.

The PVC was chosen in this project due to its low cost and good properties that help in the construction of a rigid frame. The PVC pipes house the entire mechanism and support it during the working stage. To these pipes connected are the prism shapes structure that is used to guide the garbage to the opening. Even the opening of the bin is constructed with the help of foam sheet.

To the base of the PVC frame we have attached a plastic base on which the first servo is mounted and to this servo we have attached an acrylic circle on which the containers will be kept. The containers are also constructed with the help of the foam sheet. To this PVC frame we have attached a box type structure which will house the entire circuit such as the raspberry pi 4 and the power supply and the camera which will be further adjusted on to the prism shaped guide to provide an accurate information to the pi about the detected image for classification.

The pi camera is attached to the raspberry pi and the servos are connected to the pi as well. The servos are connected to the pi with the help of jumper wires and an external power supply is given to the servo motors. A power supply is separately attached to the pi and the HDMI ports and the mouse and the keyboard are attached to it as well.

The power supply is connected to both the servos where the ground and the positive are common and the pins from where they will receive the signal is different. We can choose any pin according to our choice and then start the program.

- **Understanding the Working**

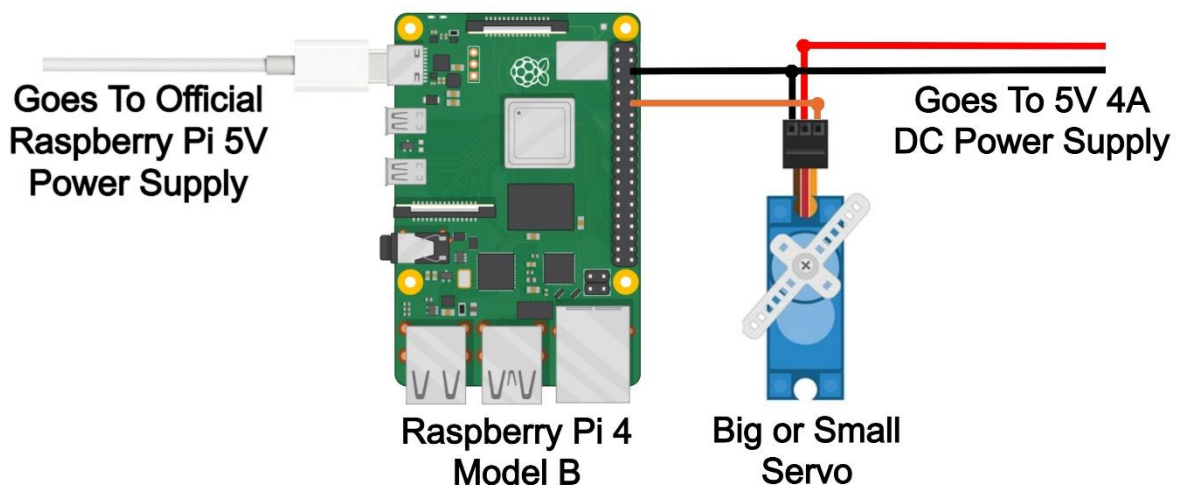
When the object is dropped in the bin the camera will detect the object and then send a signal to the raspberry pi for the classification. Once the object is classified then the pi will send the signal to the 1st servo to rotate a certain angle so that the container can be adjusted to the opening of the bin, once that is done then the opening will be opened by operating the 2nd servo and the lid opens, once that is done the garbage is dropped to the container and then the process is complete but now the container will return to its home position and the lid of the bin will close again and this process is done by giving a time delay to the pi program. The entire program is done in python and is operated by thonny.

7.IMPLEMENTAION

Implementing an AI video classifier recycle bin involves several steps, including setting up hardware components, deploying the object detection model, integrating servo control for bin sorting, and developing the application logic. Below is a high-level outline of the implementation process:

Hardware Setup:

- Connect the required hardware components, including a Raspberry Pi, PiCamera, GPIO pins for servo motors, and any additional peripherals (e.g., power supply, enclosure).
- Install necessary libraries and drivers for hardware communication and control (e.g., RPi.GPIO for GPIO control, PiCamera for camera access).

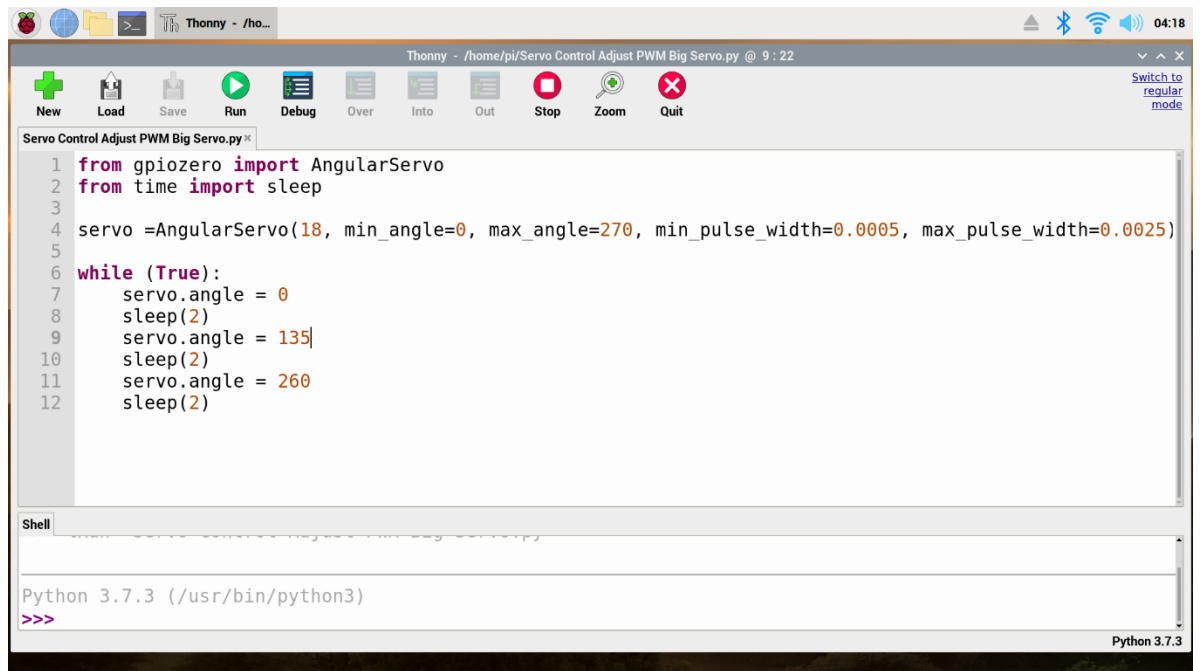


Object Detection Model:

- Choose and acquire a pre-trained object detection model suitable for classifying recyclable items (e.g., metal, paper, plastic).
- Convert the model to a format compatible with inference on the Raspberry Pi (e.g., TensorFlow Lite for optimized performance).
- Optimize the model for real-time inference on resource-constrained hardware, considering factors like model size, speed, and accuracy.

Integration with Servo Control:

- Set up servo motors to actuate the bins for sorting recyclable items.
- Define functions to control the servo motors based on detected object categories (e.g., move the servo to a specific angle for metal, paper, or plastic).



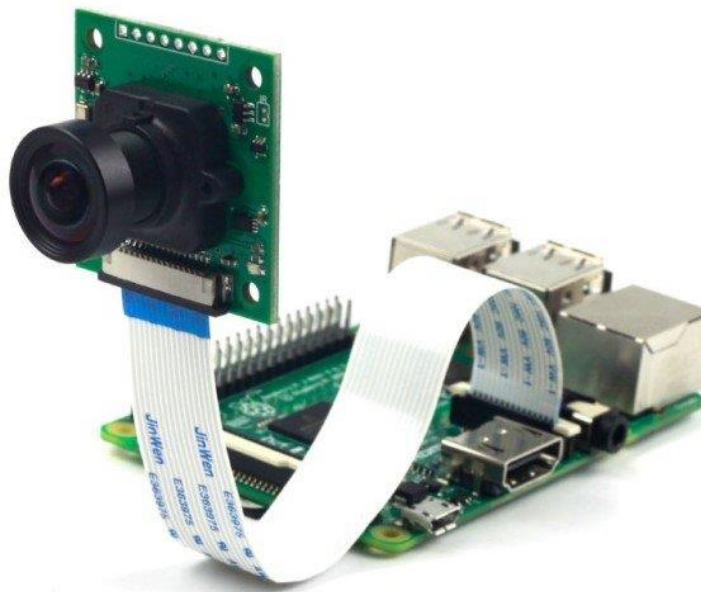
```
Thonny - /home/pi/Servo Control Adjust PWM Big Servo.py @ 9 : 22
New Load Save Run Debug Over Info Out Stop Zoom Quit
Switch to regular mode

Servo Control Adjust PWM Big Servo.py
1 from gpiozero import AngularServo
2 from time import sleep
3
4 servo = AngularServo(18, min_angle=0, max_angle=270, min_pulse_width=0.0005, max_pulse_width=0.0025)
5
6 while (True):
7     servo.angle = 0
8     sleep(2)
9     servo.angle = 135
10    sleep(2)
11    servo.angle = 260
12    sleep(2)

Shell
Python 3.7.3 (/usr/bin/python3)
>>>
```

Camera Feed Processing:

- Capture video frames from the PiCamera or external camera module.
- Preprocess the frames (e.g., resizing, color conversion) as required by the object detection model.
- Pass the processed frames to the object detection model for inference.



Object Detection and Classification:

- Run object detection on each frame using the deployed model.
- Analyze detection results to classify objects into recyclable categories (e.g., metal, paper, plastic).
- Trigger servo control actions based on the detected categories to sort items into corresponding bins.

User Interface and Feedback:

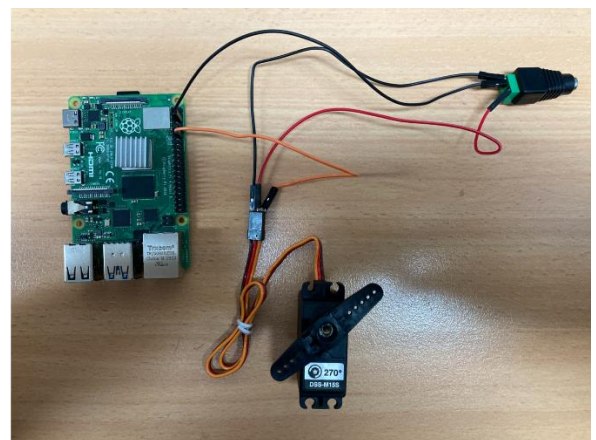
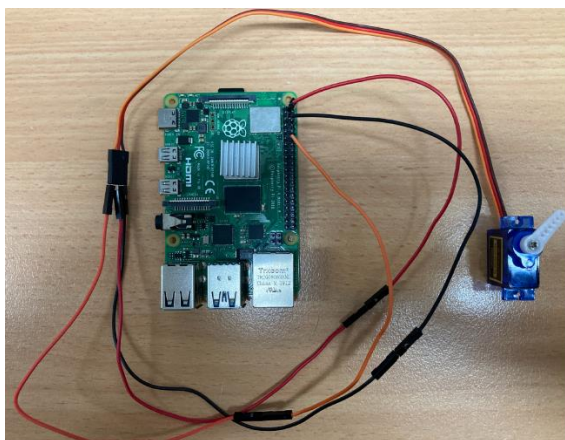
- Develop a user interface for visualizing the camera feed, detection results, and sorting actions (e.g., GUI using OpenCV).
- Provide feedback to users through visual indicators or notifications (e.g., display detected categories, sorting progress, system status).

Error Handling and Edge Cases:

- Implement error handling mechanisms to address potential issues such as network disruptions, hardware failures, or incorrect detections.
- Handle edge cases and corner scenarios to ensure robustness and reliability of the system under various conditions.

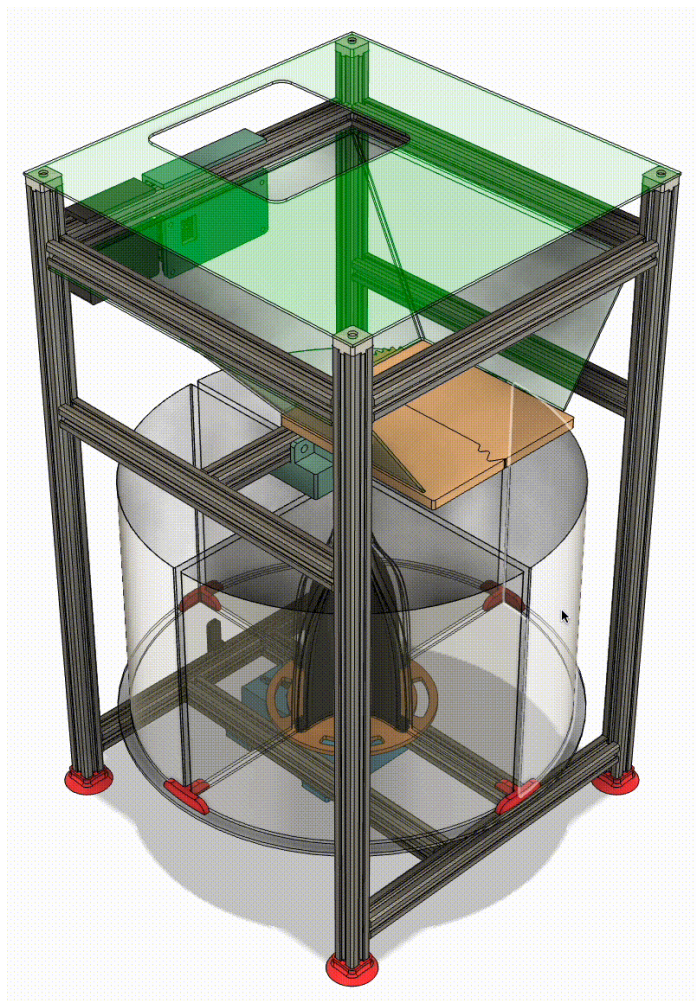
Testing and Optimization:

- Conduct thorough testing of the system to validate its functionality, performance, and accuracy.
- Optimize the system parameters (e.g., model thresholds, servo control angles) based on testing results and user feedback.
- Iterate on the design and implementation to improve efficiency, usability, and overall user experience.



Deployment and Maintenance:

- Deploy the AI video classifier recycle bin system in its target environment (e.g., public spaces, recycling facilities).
- Provide user training and documentation for operating and maintaining the system.
- Monitor system performance and conduct regular maintenance to ensure smooth operation and address any issues that arise.



8.TESTING AND TECHNICAL CHALLENGES

Testing :

An AI video classifying recycle bin involves validating its functionality, performance, and reliability under various conditions. Here are some testing considerations and potential technical challenges:

1. Functionality Testing:

- Verify that the system accurately detects and classifies recyclable items (e.g., metal, paper, plastic) in real-time.
- Test the servo control mechanism to ensure correct sorting of items into the appropriate bins based on classification results.
- Validate user interface elements such as displaying camera feed, detection results, and sorting actions accurately.

2. Performance Testing:

- Measure the processing speed of object detection and classification to ensure real-time performance, especially with the specified hardware resources.
- Evaluate the system's ability to handle a continuous stream of video frames without dropping frames or experiencing significant delays.
- Assess the overall system responsiveness and latency from object detection to sorting actions.

3. Accuracy Testing:

- Quantify the accuracy of object detection and classification by comparing the detected categories with ground truth labels for test images or videos.
- Evaluate the impact of varying environmental conditions (e.g., lighting, background clutter) on detection accuracy and reliability.
- Conduct testing with a diverse range of recyclable items to ensure robustness across different shapes, sizes, and materials.

4. Edge Cases and Error Handling:

- Test the system's behavior in edge cases and challenging scenarios, such as occluded objects, partially visible items, or ambiguous detections.
- Validate error handling mechanisms for gracefully recovering from failures or unexpected events (e.g., network disruptions, hardware malfunctions).
- Assess the system's resilience to false positives/negatives and its ability to adapt to noisy or uncertain detection outcomes.

5. Integration Testing:

- Verify seamless integration between different components of the system, including camera input, object detection module, servo control, and user interface.
- Test communication protocols and data exchange between hardware components (e.g., Raspberry Pi, servo motors) and software modules.

6. Usability Testing:

- Solicit feedback from end-users or stakeholders through usability testing sessions to evaluate the system's ease of use, intuitiveness, and user satisfaction.
- Identify any usability issues or pain points (e.g., unclear instructions, confusing interface elements) and iteratively refine the design based on feedback.

7. Scalability and Robustness Testing:

- Assess the system's scalability to handle varying loads and volumes of recyclable items, including peak usage scenarios.
- Conduct stress testing to determine the system's robustness and stability under heavy usage or adverse conditions.

8. Compliance and Safety Testing:

- Ensure compliance with relevant regulations and standards for waste management, environmental protection, and safety.
- Verify that the system adheres to privacy guidelines and data protection regulations, particularly concerning the handling of video data and personal information.

Technical Challenges:

- **Resource Constraints:** Optimizing the object detection model and servo control algorithms to run efficiently on resource-constrained hardware like the Raspberry Pi.
- **Environmental Variability:** Dealing with challenges such as changing lighting conditions, occlusions, and background clutter that can affect the accuracy of object detection.
- **Real-Time Processing:** Achieving low-latency performance for real-time processing of video streams while maintaining high detection accuracy.
- **Hardware Integration:** Ensuring seamless integration and compatibility between hardware components (e.g., camera, servos, Raspberry Pi) and software modules.
- **Robustness to Noise and Uncertainty:** Handling noisy sensor data, uncertain detections, and unexpected scenarios with robust error handling and resilience mechanisms.
- **User Interaction:** Designing a user-friendly interface that provides clear feedback and instructions while minimizing cognitive load and user errors.
- **Maintenance and Upkeep:** Addressing long-term maintenance challenges, including software updates, hardware maintenance, and system calibration to ensure continued reliability and performance.

9.CONCLUSION

In conclusion, the development of an AI video classifying recycle bin represents a promising innovation in waste management systems. By integrating artificial intelligence, computer vision, and robotic technology, this solution offers an efficient and automated approach to sorting recyclable items, contributing to environmental sustainability efforts.

Through real-time object detection and classification, the system can accurately identify various materials such as metal, paper, and plastic, enabling targeted sorting into designated bins. This not only streamlines the recycling process but also enhances the recovery of valuable resources while reducing contamination in waste streams.

Furthermore, the implementation of servo-controlled mechanisms facilitates precise sorting actions based on classification results, ensuring the proper disposal of recyclable items and minimizing waste diversion errors.

However, the development and deployment of an AI video classifying recycle bin entail various challenges, including optimizing performance on resource-constrained hardware, addressing environmental variability in detection accuracy, ensuring seamless integration of hardware and software components, and maintaining system robustness and reliability over time.

Despite these challenges, the potential benefits of AI-powered recycling solutions are substantial, offering a scalable and adaptable approach to waste sorting that aligns with the principles of circular economy and sustainable resource management. With further research, innovation, and collaboration across industries, AI video classifying recycle bins hold promise for revolutionizing waste management practices and fostering a cleaner, greener future for generations to come.

In the ongoing quest for sustainable waste management solutions, the emergence of AI video classifying recycle bins represents a transformative innovation with far-reaching implications. By harnessing the power of artificial intelligence, computer vision, and robotic technology, these systems offer a novel approach to streamlining waste sorting processes and enhancing recycling efficiency.

At the core of this innovation lies the ability to autonomously identify and classify recyclable materials in real-time. Through advanced object detection algorithms, AI video classifying recycle bins can accurately distinguish between various types of materials, including metal, paper, plastic, and more. This capability not only facilitates precise sorting but also enables the diversion of valuable resources from landfills back into the recycling stream.

Moreover, the integration of servo-controlled mechanisms enables these bins to execute sorting actions with precision and accuracy. By directing items to designated bins based on their material composition, AI video classifying recycle bins minimize contamination and optimize resource recovery, thereby contributing to a more sustainable and circular economy.

However, the development and deployment of these systems are not without challenges. Technical hurdles such as optimizing performance on resource-constrained hardware, addressing environmental variability in detection accuracy, and ensuring seamless integration of hardware and software components require careful consideration and ongoing refinement.

Despite these challenges, the potential benefits of AI video classifying recycle bins are immense. By automating and optimizing waste sorting processes, these systems can reduce reliance on manual labor, improve recycling rates, and mitigate environmental impacts associated with improper waste disposal. Moreover, they pave the way for innovative approaches to waste management that prioritize efficiency, sustainability, and environmental stewardship.

In conclusion, AI video classifying recycle bins represent a significant step forward in the quest for sustainable waste management solutions. With continued research, innovation, and collaboration, these systems have the potential to revolutionize recycling practices and usher in a cleaner, greener future for generations to come.

10.REFERENCES

- <https://www.youtube.com/watch?v=3YqbO2AlepM>
- <https://github.com/freedomwebtech/tflite-custom-object-bookworm>
- <https://core-electronics.com.au/guides/control-servo-raspberry-pi/>
- <https://www.youtube.com/watch?v=40tZQPd3z8g>
- <https://www.youtube.com/watch?v=40tZQPd3z8g>
- <https://www.hackster.io/sarful/how-to-interface-servo-motor-with-raspberry-pi-a2d8dc>
- <https://hackaday.io/project/191802-ai-audio-classifier-recycle-bin>
- <https://www.youtube.com/watch?v=jqvCsm-7A4>
- <https://www.youtube.com/watch?v=roWY29RNFU0>
- <https://projects.raspberrypi.org/en/projects/grandpa-scarer/3>
- <https://www.instructables.com/Servo-Motor-Control-With-Raspberry-Pi/>
- <https://colab.research.google.com/>